

IBM Data Engineer Assessment Roadmap

COMPLETE EXECUTION BLUEPRINT & MEMORY CHEAT SHEET

Preparation Milestones & Targets

100 SQL PROBLEMS	100 PYTHON TASKS	75 DSA CORE	5 E2E DE PROJECTS	10 MOCK INTERVIEWS
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1. Coding & Data Structures (DSA) Checklist

Category	Topic / Problem	LeetCode (Clickable)	HackerRank Equivalent
Arrays	☐ Two Sum	1: Two Sum	Left Rotation
	☐ Best Time to Buy & Sell Stock	121: Stock I	Array Manipulation
	☐ Maximum Subarray	53: Max Subarray	Max Subarray
	☐ Merge Sorted Array	88: Merge Sorted	Dynamic Array
	☐ Move Zeroes	283: Move Zeroes	Sparse Arrays
	☐ Product of Array Except Self	238: Product Except Self	Arrays Domain
Strings	☐ Reverse String	344: Rev String	String Reverse
	☐ Reverse Words in a String	151: Rev Words	Strings Core
	☐ Valid Palindrome	125: Palindrome	String Validators
	☐ Valid Anagram	242: Anagram	Making Anagrams
	☐ Longest Substring W/O Repetition	3: Longest Substr	Alternating Char
	☐ Longest Common Prefix	14: Prefix	Caesar Cipher
	☐ First Unique Character	387: Unique Char	Validators
HashMap	☐ Contains Duplicate	217: Duplicate	Dictionaries & Maps
	☐ Group Anagrams	49: Group Anagram	Hash Tables Core
	☐ Majority Element	169: Majority El	Frequency Queries
	☐ Top K Frequent Elements	347: Top K	Collections.Counter
Stack / Queue	☐ Valid Parentheses	20: Parentheses	Balanced Brackets
	☐ Min Stack	155: Min Stack	Text Editor
	☐ Queue using Stacks	232: Queue Stack	Queue via 2 Stacks
Linked List	☐ Reverse Linked List	206: Reverse LL	Reverse LL
	☐ Merge Two Sorted Lists	21: Merge LL	Insert Node
	☐ Linked List Cycle	141: LL Cycle	Cycle Detection
Trees & Search	☐ Maximum Depth of Binary Tree	104: Max Depth	Tree Height
	☐ Binary Search	704: Bin Search	BST Search

Category	Topic / Problem	LeetCode (Clickable)	HackerRank Equivalent
	☐ Merge Intervals	56: Intervals	Big Sorting
	☐ Number of Islands	200: Islands	BFS Short Reach

2. SQL Mastery Track (Highest Priority)

SQL Focus Topic / Pattern	LeetCode ID	Direct Link
☐ Combine Two Tables (Basic Joins)	175	175: Link
☐ Second Highest Salary (Subqueries / Offset)	176	176: Link
☐ Duplicate Emails (Aggregation + Having)	182	182: Link
☐ Customers Who Never Order (Anti-Joins)	183	183: Link
☐ Employees Earning More Than Managers (Self Joins)	181	181: Link
☐ Department Highest Salary (Window / Group By)	184	184: Link
☐ Department Top Three Salaries (Dense Rank)	185	185: Link
☐ Rank Scores (Window Functions)	178	178: Link
☐ Rising Temperature (Date Math + Self Join)	197	197: Link
☐ Managers with ≥ 5 Reports (Having / Filtering)	570	570: Link
☐ Human Traffic of Stadium (Sequential Data Windowing)	601	601: Link
☐ Trips and Users (Complex Filters & Joins)	262	262: Link

Must-Master Patterns: Window Functions (`ROW_NUMBER()` , `RANK()` , `DENSE_RANK()`), CTEs (Common Table Expressions), Recursive Queries, and Advanced Partitioning.

3. Python Fundamentals & Scripting

Topic	Platform Target	Key Engineering Standard
☐ Reverse & Filter Strings	LC 344	Avoid extra memory overhead; slice vs mutation.
☐ Fibonacci & Prime Numbers	LC 509 / HR Prime	Understand recursion depth and Memoization (<code>@lru_cache</code>).
☐ Frequency Dictionaries	LC 347	Use <code>collections.Counter</code> for O(N) execution.
☐ Flatten Nested Lists	LC 341	Yield items using generators to preserve streaming.
☐ File I/O & CSV Handling	Python Built-in / HR	Always use contexts managers (<code>with open(...)</code>).
☐ Decorators & Generators	HackerRank Core	Memory efficient streaming using <code>yield</code> instead of full arrays.

Topic	Platform Target	Key Engineering Standard
☐ Functional Paradigms	Map / Filter / Reduce	Know the <code>functools.reduce</code> syntax for aggregations.

High-Yield Data Engineering Cheat Sheet

QUICK-RECALL MEMORY REFERENCE FOR INTERVIEWS

1. ETL vs ELT Architecture

ETL (Extract, Transform, Load): Transformation occurs before data hits target destination. Best for high-security, structural compliance, or when legacy data warehouses are targets. High computing requirements on the processing engine.

ELT (Extract, Load, Transform): Data loaded raw into Cloud Data Lakes/Warehouses; transformations happen on-demand using target compute (e.g., Snowflake, BigQuery). Faster ingestion, highly scalable.

2. Data Modeling: Star vs. Snowflake Schema

Star Schema: Fact table surrounded by denormalized Dimension tables. Simpler queries, fewer JOINS, faster read optimization, but data redundancy is higher.

Snowflake Schema: Dimension tables are normalized into further lookup tables. Minimizes data redundancy, saves space, but increases complexity due to multi-layer JOINS.

3. Processing: Batch vs. Streaming

Batch: Processes bounded, historical blocks of data (e.g., Daily Apache Spark jobs, Airflow workflows). Highly optimized for massive volume throughput.

Streaming: Processes unbounded, real-time message streams item-by-item or via micro-batches (e.g., Apache Kafka, Spark Streaming). Optimized for low-latency insight.

4. Distributed Storage & Compute Performance

Partitioning: Physically splitting data directory trees by a column key (e.g., `year/month/day`). Prevents full-table scans.

Data Skew: Occurs when partition keys are unevenly distributed, forcing one cluster node to process 90% of the data while others stay idle. Fix via *salting* (appending random prefixes to keys).

5. Core Distributed Theorems

CAP Theorem: A distributed system can guarantee at most two out of three: **C**onsistency (same data everywhere), **A**vailability (every request gets a response), **P**artition Tolerance (system functions despite network drops). Interconnected with ACID vs BASE properties.

ACID Properties: Atomicity, Consistency, Isolation, Durability. Ensures transactional integrity (vital for OLTP systems).

6. Big Data Toolkit Cheat Sheet

Apache Spark: In-memory distributed processing framework. Uses Lazy Evaluation (DAGs aren't run until an Action like `.collect()` or `.save()` is called).

Apache Kafka: Distributed event-streaming platform using append-only logs. Topics are split into Partitions for parallel consumption.

Apache Airflow: Orchestration tool that manages pipelines as Directed Acyclic Graphs (DAGs). Does not move data itself; merely schedules and triggers tasks.